Algorithms 2

Approximating the Maximum Cut Problem

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1. The Max Cut Problem

Problem 1 Max Cut

Given: An undirected graph $G = (V, E)$.

Find: A bipartition $(S, V \setminus S)$ maximising $e_G(S, V \setminus S)$ – the number of edges crossing the cut.

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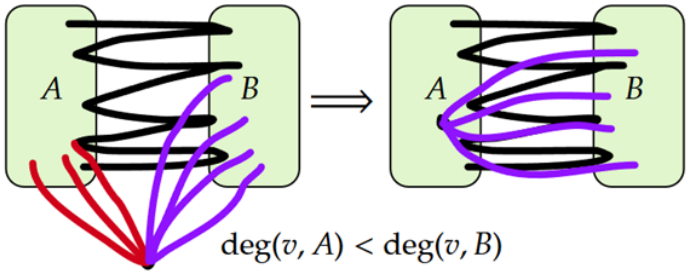
Theorem 3 For every graph G: max-cut$(G) \geq \frac{e(G)}{2}$.

Observation 3 Equivalently: every graph G has a **spanning bipartite subgraph** with $\geq \frac{e(G)}{2}$ edges.

Proof: By induction on $v(G)$. Base cases $v(G) = 1, 2$ are trivial.

Inductive step: Let $v \in V(G)$ be arbitrary.

- Apply the IH to $G - v$:
  - obtain a bipartition $(S, V(G) \setminus (S \cup \{v\}))$ with $\geq \frac{e(G-v)}{2}$ crossing edges.
- Place v on the side where it has the **fewest** neighbours in G
- Then v contributes at least $\deg_G(v)/2$ crossing edges.



Size of resulting cut in G:

$$\geq \frac{e(G-v)}{2} + \deg_G(v)/2 = \frac{e(G)}{2}. \quad \square$$

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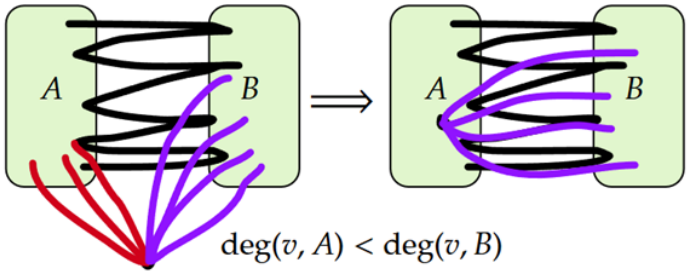

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Theorem 5 The above proof yields a poly-time algorithm forming a $\frac{1}{2}$-approximation for Max Cut.

2. A Randomised Construction

Algorithm 1: Rand-Cut

1: $A, B \leftarrow \emptyset$
2: **for** each $v \in V(G)$ **do**
3: Toss a fair coin independently
4: **Heads:** place v in A. **Tails:** place v in B.
5: **end**
6: **return** (A, B)

Algorithm 2: Rand-Cut

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Observation 5 The randomised algorithm achieves an **expected** cut of size $\frac{e(G)}{2}$.

For each edge $e = uv \in E(G)$, define the indicator:

$$X_e := \begin{cases} 1 & e \in E_G(A,B) \\ 0 & e \notin E_G(A,B) \end{cases}$$

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Since the coins are **independent**:

$$\mathbb{E}[X_e] = \mathbb{P}[u \in A \wedge v \in B] + \mathbb{P}[u \in B \wedge v \in A] = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

We may write $e_G(A,B) = \sum_{e \in E(G)} X_e$, so by **linearity of expectation**:

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But how often do we actually get a cut $\geq \frac{e(G)}{2}$?

Theorem 6 The algorithm produces a cut of size $\geq \frac{e(G)}{2}$ with probability $\geq \frac{1}{e(G)+1}$. **(This is bad!)**

proof:

- Let $p := \mathbb{P}\left[e_G(A,B) \geq \frac{e(G)}{2}\right]$.
  - **Goal** bound p from below.

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proof:

- Let $p := \mathbb{P}\left[e_G(A,B) \geq \frac{e(G)}{2}\right]$.
  - **Goal** bound p from below.

Write the expectation by splitting on the cut value:

$$\begin{aligned}\frac{e(G)}{2} = \mathbb{E}[e_G(A,B)] &= \sum_{i \leq \frac{e(G)}{2}-1} i \cdot \mathbb{P}[e_G(A,B)=i] + \sum_{i \geq \frac{e(G)}{2}} i \cdot \mathbb{P}[e_G(A,B)=i] \\ &\leq \left(\frac{e(G)}{2}-1\right) \cdot \underbrace{\sum_{i \leq \frac{e(G)}{2}-1} \mathbb{P}[e_G(A,B)=i]}_{1-p} + e(G) \cdot \underbrace{\sum_{i \geq \frac{e(G)}{2}} \mathbb{P}[e_G(A,B)=i]}_{p}\end{aligned}$$

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Rearranging:

$$\frac{e(G)}{2} \leq \left(\frac{e(G)}{2}-1\right)(1-p) + e(G) \cdot p$$

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- $p := \mathbb{P}\left[e_G(A,B) \geq \frac{e(G)}{2}\right]$
- $\frac{e(G)}{2} \leq \left(\frac{e(G)}{2} - 1\right)(1-p) + e(G) \cdot p$

Isolating p:

$$\frac{e(G)}{2} \leq \frac{e(G)}{2} - 1 - p \cdot \left(\frac{e(G)}{2} - 1\right) + e(G) \cdot p$$

$$1 \leq p \cdot \left(e(G) - \frac{e(G)}{2} + 1\right) = p \cdot \left(\frac{e(G)}{2} + 1\right)$$

$$\implies \quad p \geq \frac{1}{\frac{e(G)}{2} + 1} \geq \frac{1}{e(G)+1}. \quad \square$$

3. De-randomisation

3.1 Conditional Expectation Background

Goal: Turn the randomised algorithm into a **deterministic** one with the same guarantee.

Key tool: the **method of conditional expectations**.

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Definition 5 Conditional Expectation

For discrete RVs X, Y and value z in the range of Y:

$$\mathbb{E}[X \mid Y = z] := \sum_x x \cdot \mathbb{P}[X = x \mid Y = z]$$

We write $\mathbb{E}[X \mid Y]$ for the **function** $z \mapsto \mathbb{E}[X \mid Y = z]$.

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Theorem 12 Law of Total Expectation

For discrete RVs X, Y:

$$\mathbb{E}[X] = \sum_{z \in \text{ range}(Y)} \mathbb{P}[Y = z] \cdot \mathbb{E}[X \mid Y = z]$$

Let $v_1, ..., v_n$ be an ordering of the vertices. Suppose $v_1, ..., v_i$ have already been placed in A or B.

By the **Law of Total Expectation**, for vertex v_{i+1}:

$$\mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}] = \frac{1}{2} \cdot \mathbb{E}[e_G(A,B) \mid ..., v_{i+1} \in A] + \frac{1}{2} \cdot \mathbb{E}[e_G(A,B) \mid ..., v_{i+1} \in B]$$

Let:

- $\mathbb{E}_A = \mathbb{E}[e_G(A,B) \mid ..., v_{i+1} \in A]$
- $\mathbb{E}_B = \mathbb{E}[e_G(A,B) \mid ..., v_{i+1} \in B]$

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So:

$$2\mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}] = \mathbb{E}_A + \mathbb{E}_B$$

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So:

$$2\mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}] = \mathbb{E}_A + \mathbb{E}_B$$

Observation 8 Either $\mathbb{E}_A \geq \mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}]$ or $\mathbb{E}_B \geq \mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}]$.

4. The Deterministic Algorithm

Assuming: $\mathbb{E}_A$ and $\mathbb{E}_B$ can be computed efficiently we have the following algorithm.

Algorithm 3: Derand-Max-Cut

```
v_1, ..., v_n ← an arbitrary ordering of V
A, B ← ∅
for i = 1 to n do
    E_A ← Compute 𝔼[e_G(A, B) | v_1, ..., v_{i-1} placed ∧ v_i ∈ A]
    E_B ← Compute 𝔼[e_G(A, B) | v_1, ..., v_{i-1} placed ∧ v_i ∈ B]
    If E_A ≥ E_B place v_i in A, else place v_i in B
end
return (A, B)
```

Theorem 13 Derand-Max-Cut is a deterministic poly-time $\frac{1}{2}$-approximation for Max Cut.

Proof (By induction):

$$\text{I.H. } \forall i < k, \mathbb{E}[e_G(A,B) \mid v_1, ..., v_i \text{ placed}] \geq \frac{e(G)}{2}$$

Assume without loss of generality that $\mathbb{E}_A \geq \mathbb{E}_B$.

- By assumption we have $\mathbb{E}_A \geq \mathbb{E}[e_G(A,B) \mid v_1, ..., v_{k-1} \text{ placed}] \geq \frac{e(G)}{2}$.

$\implies$ for $k = n$ we have $e_G(A,B) = \mathbb{E}[e_G(A,B) \mid v_1, ..., v_n \text{ placed}] \geq \frac{e(G)}{2}$ as required.

For each step i, we need to compute efficiently:

$$\mathbb{E}_A := \mathbb{E}\big[e_G(A,B) \mid v_1,\ldots,v_i \text{ placed} \wedge v_{i+1} \in A\big]$$

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Let $\tilde{A}, \tilde{B}$ be the **current actual sets** with $v_1,\dots,v_i$ placed. Define:

$$\ell := e(G) - \deg_G\left(v_{i+1}, \tilde{A}\right) - \deg_G\left(v_{i+1}, \tilde{B}\right) - e_G\left(\tilde{A}\right) - e_G\left(\tilde{B}\right) - e_G\left(\tilde{A}, \tilde{B}\right)$$

$\ell = \#$ edges among the **remaining** vertices $v_{i+2},\dots,v_n$ (still random).

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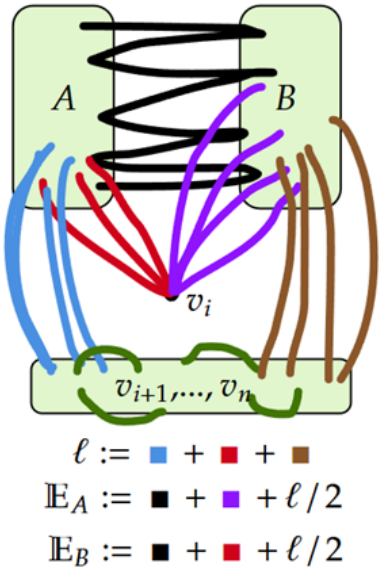
$\ell = \#$ edges among the **remaining** vertices $v_{i+2},\dots,v_n$ (still random).

Then:

$$\mathbb{E}_A = e_G\left(\tilde{A}, \tilde{B}\right) + \deg_G\left(v_{i+1}, \tilde{B}\right) + \frac{\ell}{2}$$

$$\mathbb{E}_B = e_G\left(\tilde{A}, \tilde{B}\right) + \deg_G\left(v_{i+1}, \tilde{A}\right) + \frac{\ell}{2}$$



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$\ell = \#$ edges among the **remaining** vertices $v_{i+2},\dots,v_n$ (still random).

Then:

$$\mathbb{E}_A = e_G\left(\tilde{A}, \tilde{B}\right) + \deg_G\left(v_{i+1}, \tilde{B}\right) + \frac{\ell}{2}$$

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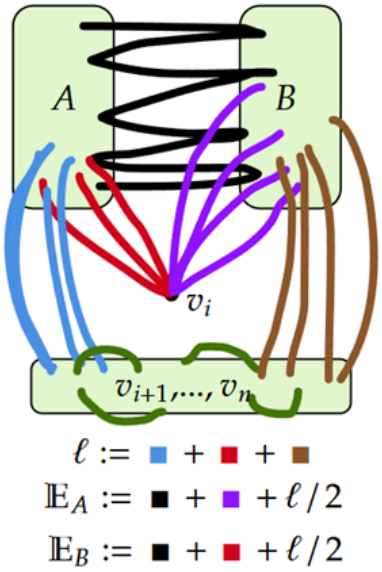
Therefore:

$$|\mathbb{E}_A - \mathbb{E}_B| = |\deg_G\left(v_{i+1}, \tilde{B}\right) - \deg_G\left(v_{i+1}, \tilde{A}\right)|$$



$$|\mathbb{E}_A - \mathbb{E}_B| = |\deg_G(v_{i+1}, \tilde{B}) - \deg_G(v_{i+1}, \tilde{A})|$$

Therefore:

$$\mathbb{E}_A \geq \mathbb{E}_B \iff \deg_G(v_{i+1}, \tilde{B}) \geq \deg_G(v_{i+1}, \tilde{A})$$

$$|\mathbb{E}_A - \mathbb{E}_B| = |\deg_G(v_{i+1}, \tilde{B}) - \deg_G(v_{i+1}, \tilde{A})|$$

Therefore: $$\mathbb{E}_A \geq \mathbb{E}_B \iff \deg_G(v_{i+1}, \tilde{B}) \geq \deg_G(v_{i+1}, \tilde{A})$$

So we may write:

Algorithm 5: Derand-Max-Cut

1: $v_1, ..., v_n \leftarrow$ an arbitrary ordering of V
2: $A, B \leftarrow \emptyset$
3: **for** $i = 1$ to n **do**
4: | **If** $\deg_G(v_{i+1}, B) \geq \deg_G(v_{i+1}, A)$ place v_i in A, **else** place v_i in B
5: **end**
6: **return** (A, B)

This is the first "algorithm" we saw in those slides.